

HEALTHCARE PERFORMANCE DASHBOARD

Patient, Clinical, Doctor Performance & Financial Analytics

PROJECT REPORT

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Microsoft Power BI Portfolio Project

1. Project Overview

The Healthcare Performance Dashboard is an interactive Microsoft Power BI analytics project designed to provide a consolidated view of patient activity, clinical patterns, doctor performance, billing, health insurance coverage, and follow-up information. The dashboard converts patient-level healthcare data into a visual management and analysis tool that can be used to identify patterns, compare performance, and support data-driven decision-making.

The dashboard is designed as a single-page analytical report with interactive filtering. Users can explore the information by patient and admission date range while the main visuals respond to the selected filters.

2. Project Title

Healthcare Performance & Patient Analytics Dashboard

Dashboard title used in Power BI: "Healthcare Performance Dashboard".

3. Business Problem / Purpose

Healthcare operational data can contain information about admissions, discharge activity, diagnoses, tests, doctors, patient feedback, billing, insurance and bed usage. When this information is reviewed only through raw tables, it becomes difficult to identify important patterns quickly.

This project addresses that problem by creating a visual analytics layer that helps users:

- Monitor important patient and financial indicators from one dashboard.
- Understand how patients are distributed across bed categories.
- Compare doctor-level revenue, insurance amounts, feedback and patient volume.
- Identify high-volume diagnoses and understand the relative patient load.
- Compare billed amounts with health insurance amounts across diagnostic tests.
- Filter the dashboard by patient and admission date range for focused analysis.
- Use interactive visual analysis instead of depending only on manual spreadsheet reporting.

4. Project Objectives

- Build an interactive healthcare analytics dashboard using Microsoft Power BI.
- Transform patient-level data into clear KPIs and business-friendly visualizations.
- Provide visibility into patient activity and important date information.
- Analyse diagnosis and bed-category distribution.
- Evaluate doctor-level operational, feedback and financial performance.
- Compare billing and health insurance amounts by diagnostic test.
- Create a portfolio-quality dashboard demonstrating practical Power BI and analytical skills.

5. Data Used in the Project

The Power BI report is built around a structured patient-level healthcare dataset. The report model uses the Sheet1 dataset and the dashboard queries patient, admission, clinical, doctor, feedback and financial fields.

Data Area	Fields / Metrics Used	Analytical Purpose
Patient	Patient_ID	Patient-level filtering and patient volume analysis
Admission / Discharge	Admit_Date, Discharge_Date, Followup Date	Timeline and follow-up visibility
Clinical	Diagnosis, Test	Diagnosis volume and test-level financial analysis
Bed / Capacity	Bed_Occupancy	Patient distribution by bed category
Doctor	Doctor	Doctor-level comparison and performance analysis
Feedback	Feedback	Doctor-level average feedback analysis
Financial	Billing Amount	Revenue / billing analysis
Insurance	Health Insurance Amount	Insurance amount analysis and comparison with billing

The dashboard includes an admission-date range covering 25 January 2023 to 3 March 2024 in the report's configured filtering context. The exact values displayed by the KPI cards change according to the active dashboard filters.

6. Data Preparation & Power BI Approach

The project follows a practical Power BI reporting workflow. The healthcare dataset is brought into Power BI, structured for reporting, and then used across multiple visuals with aggregation and filtering.

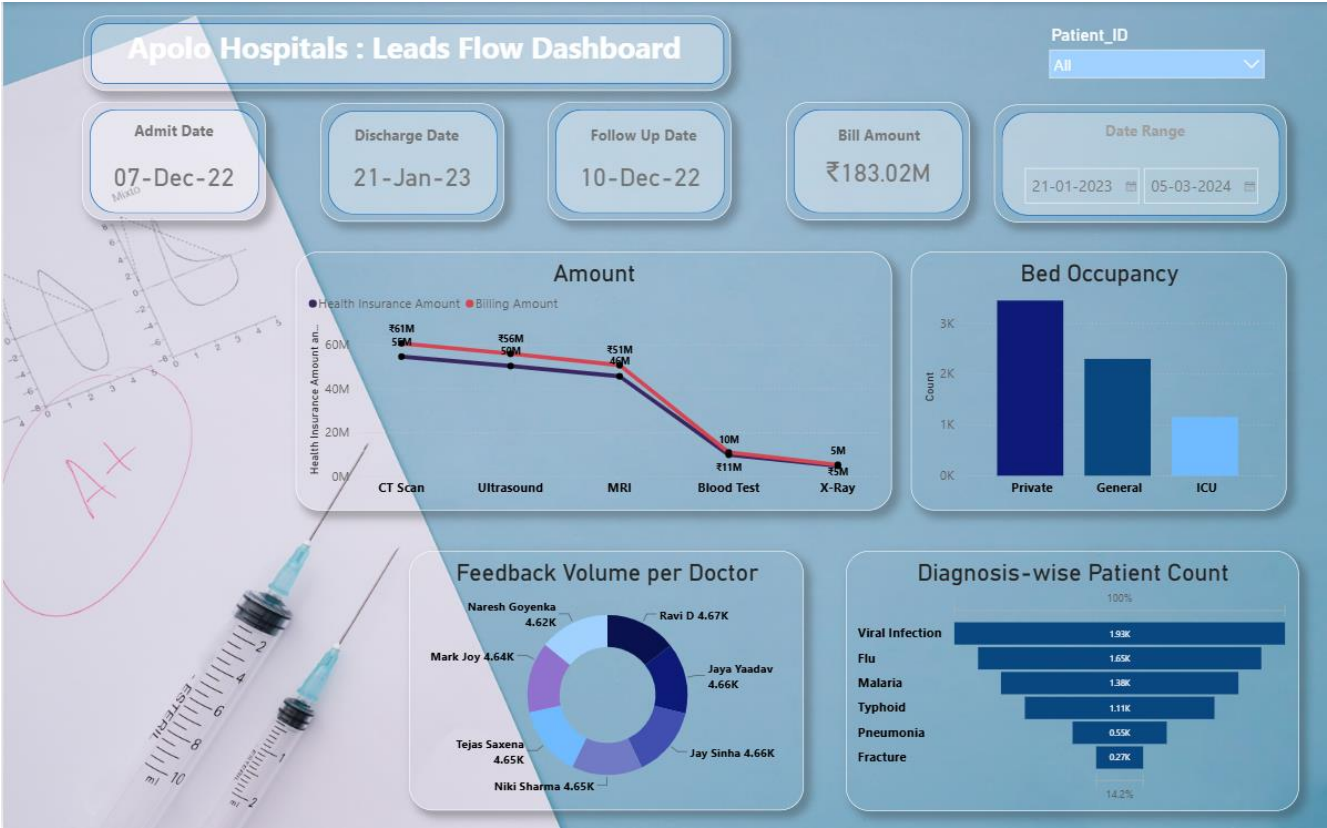
- Imported and structured the healthcare dataset in Power BI.
- Reviewed fields used for patient, clinical, doctor and financial analysis.
- Used date fields for admission, discharge and follow-up analysis.
- Applied aggregation such as count, sum and average according to the business question.
- Configured interactive slicers for Patient_ID and Admit_Date.
- Configured visual interactions so selections can be used to analyse the dashboard dynamically.
- Applied sorting, titles, formatting, cards, shapes and visual styling to improve dashboard readability.

7. Dashboard KPI Layer

The upper section of the dashboard provides quick-reference KPI information before the user moves into detailed charts.

KPI	Power BI Calculation Logic	Business Meaning
Earliest Admission	Minimum Admit_Date in the active filter context.	Shows the beginning of the selected admission period.
Latest Discharge	Minimum Discharge_Date in the configured visual context.	Provides a discharge-date reference for the report.
Follow-Up Scheduled	Minimum Followup Date in the configured visual context.	Provides visibility into follow-up scheduling.
Bill Amount	Sum of Billing Amount.	Provides a consolidated billing value for the selected data.

8. Dashboard Visualizations – Detailed Explanation



Visual	Type	Fields / Measures	Purpose
1. Patient Distribution by Bed Category	Column chart	Bed_Occupancy + Count of Bed_Occupancy	Shows the number of patient records across different bed categories. It helps understand the distribution of patients across categories such as General, ICU and Private.
2. Patient Consultations by Attending Doctor	Donut chart	Doctor + Sum of Feedback	Provides a doctor-level view using feedback aggregation. The visual helps compare the contribution of each attending doctor within the selected dataset.
3. Top Diagnoses by Patient Volume	Funnel chart	Diagnosis + Count of Diagnosis	Ranks diagnoses according to patient volume. This makes it easier to identify higher-volume clinical categories and compare them with lower-volume categories.
4. Billed vs. Insured Amount by Diagnostic Test	Line chart	Test + Sum of Health Insurance Amount + Sum of Billing Amount	Compares the financial value associated with diagnostic tests. The visual highlights the relationship between total billed amount and health insurance amount across tests.
5. Revenue & Insurance by Doctor	Clustered bar chart	Doctor + Sum of Billing Amount + Sum of Health Insurance Amount	Compares doctor-wise billing and insurance amounts side by side. This provides a direct financial comparison across doctors.
6. Performance by Doctor	Clustered bar chart	Doctor + Average Feedback + Count of Patient_ID	Combines doctor-level feedback and patient volume.

			It helps compare service feedback with the number of patients associated with each doctor.
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9. Interactivity & Filtering

Interactivity is an important part of the dashboard design. Instead of presenting only static totals, the report allows the user to narrow the analysis and review the same KPIs and charts for a selected subset.

- Patient_ID slicer for patient-level exploration.
- Admission date range slicer for time-period analysis.
- Cross-filtering between dashboard visuals to support exploratory analysis.
- Dynamic KPI and chart values based on the active filter context.
- Visual titles and structured layout designed for quick business interpretation.

10. Analytical Insights the Dashboard Is Designed to Provide

The dashboard is primarily an analytical tool, so the following are the types of insights it enables the user to identify from the underlying data. They should be interpreted for the selected filter context.

- Which bed category has the highest patient volume.
- Which diagnoses represent the largest patient volumes.
- Which doctors handle higher patient volumes.
- How doctor-level average feedback compares across the selected population.
- Which doctors generate higher billing amounts and how their insurance amounts compare.
- Which diagnostic tests have comparatively higher billed amounts.
- How health insurance amounts compare with billed amounts across diagnostic tests.
- How the analysis changes when the user selects a particular patient or admission-date period.

11. Project Outcomes

The project converts a healthcare dataset into an interactive management-style dashboard. The main outcome is a single analytical view that brings together patient, clinical, operational, doctor-performance and financial information.

- Created a complete single-page healthcare performance dashboard in Microsoft Power BI.
- Converted multiple healthcare fields into meaningful KPIs and visual analytics.
- Enabled patient and date-based filtering for focused analysis.
- Provided doctor-level financial and performance comparison.
- Provided diagnosis and bed-category volume analysis.
- Added billing-versus-insurance analysis by diagnostic test.
- Improved the readability and usability of raw healthcare data through visual storytelling.
- Created a portfolio project demonstrating practical Power BI dashboard development and business analysis.

12. Tools & Skills Demonstrated

- Microsoft Power BI
- Data visualization and dashboard design
- Data aggregation and analytical reporting
- KPI development
- Interactive slicers and filtering
- Healthcare / patient analytics
- Financial and insurance analysis
- Doctor performance analysis
- Clinical volume analysis
- Business-oriented data storytelling

13. Analytical Methodology

Stage	Approach
Understand	Reviewed the available healthcare fields and identified the main analytical areas: patients, dates, diagnoses, tests, doctors, feedback, billing, insurance and bed categories.
Structure	Organised the dataset and selected fields required for patient, clinical, operational and financial analysis.
Measure	Used counts, sums and averages based on the analytical requirement of each visual.
Visualise	Selected column, donut, funnel, line and clustered bar charts according to the type of comparison required.
Interact	Added Patient_ID and admission-date filtering so users can move from overall analysis to a focused view.
Interpret	Designed the dashboard so the user can compare volumes, financial values and doctor performance from one page.

14. Portfolio Project Description

A concise portfolio description that can be used with the project:

Developed an interactive Healthcare Performance Dashboard in Microsoft Power BI to analyse patient distribution, diagnoses, doctor performance, feedback, billing and health insurance data. Built KPI cards, interactive Patient_ID and admission-date filters, and multiple analytical visuals including bed-category distribution, diagnosis volume, doctor-wise revenue and insurance comparison, test-wise billing analysis, and doctor performance analysis. The project demonstrates practical skills in data visualisation, KPI reporting, interactive dashboard design and business-focused healthcare analytics.

15. Limitations & Scope for Future Enhancement

The current dashboard focuses on the fields and visuals available in the project dataset. The following enhancements could be considered in a future version:

- Add a dedicated Date table for more advanced time-intelligence analysis.
- Add admission-type, gender, age-group, medication and hospital-level analysis if those fields are available.

- Add calculated measures such as average length of stay, insurance coverage percentage and revenue per patient.
- Add conditional formatting or benchmark indicators for doctor performance.
- Add a dedicated executive summary page and a detailed patient/clinical drill-through page.
- Add data-quality checks and refresh documentation for a production reporting environment.

16. My Role in the Project

As the developer and analyst of this portfolio project, I was responsible for understanding the healthcare dataset, deciding the key analytical questions, selecting appropriate Power BI visuals, building the dashboard layout, configuring filters and aggregations, and presenting the analysis in a business-friendly format.

- Data understanding and analytical requirement identification.
- Power BI report and dashboard development.
- KPI and visual selection.
- Patient, clinical, doctor and financial analysis.
- Interactive slicer and filter configuration.
- Dashboard formatting and visual storytelling.
- Interpretation of analytical outputs for portfolio presentation.

17. Conclusion

The Healthcare Performance Dashboard demonstrates how Microsoft Power BI can be used to convert healthcare data into a practical analytical solution. By combining KPI cards, interactive filters and multiple visual perspectives, the dashboard provides a clear way to explore patient activity, clinical patterns, doctor performance and financial information.

The project reflects a business-oriented approach to data analytics: first understand the data and business questions, then select the right measures and visualisations, and finally present the results in a format that supports faster and clearer decision-making.